

LOADZONE · CALCULATION NOTE

LoadZone sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Airtime, latency, duty cycle and link rerun for US915 with the 11-byte payload (LDZ-DEC-001 item 2); antenna line 4 at 915 MHz; FieldNode priced at USD 139.50; no requirement status changed
0.3	2026-10-01	Amish Chadha	Constructable design (LDZ-DDR-003) re-run; cost reported against the value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (airtime, energy and cell life, latency, link, detection, loads, size and mass, sign option, cost) with a status for every requirement

LoadZone sizing calculations

On paper, LoadZone meets eleven of its fifteen requirements (seven by calculation, four by design), has three at risk and leaves one that only an installation trial can settle; none is now not met. Version 0.2 applies Amish's decisions of 2026-09-25 (LDZ-DDR-002). **R4**, restated as a gateway within 300 m of the bay, is met on paper: from road level under a parked van a puck reaches about 342 m in a street canyon at SF9 with a 10 dB fade margin (it was not met against the TRL 2 target of 1 km). **R11**, restated as daylight only, is met on paper at about 28 m (it was not met at night). R1 (detection), R5 (airtime at slow data rates) and R6 (bond under braking, now with a two-part epoxy bed) are at risk. The core numbers of the TRL 2 concept stand or improve: cell life is 7.7 years at SF9, the worst-case report time is 18.2 s, and the two-puck kit is estimated at USD 116.00 against a value-engineering target of USD 120. Version 0.3 re-ran every figure on the constructable design of LDZ-DDR-003 (base tray, cradles, moved cells and antenna, channel brackets on the sign): mass, loads, link and energy were unchanged, and only the cost moved. Version 0.4 applies Amish's decision of 2026-10-02 that US915 is the default band (LDZ-DEC-001, item 2): airtime, latency and the duty-cycle rules are rerun for US915 with the 11-byte payload, and the link at 915 MHz. No requirement changes status; the worst-case report time falls from 38.6 s to 18.2 s because US915 has no duty-cycle wait, and the range at SF9 falls from 353 m to 342 m.

Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B2], is the line of the script's output that carries it. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`. The script imports `PARAMS`, `derived()` and the part volumes from `cad/src/model.py`, so the geometry here is the geometry in the STEP files and in drawing LDZ-DWG-001. It reads the costs from `bom/bom.csv` and `budget_usd` from `project.yaml`, which is a hypothetical value-engineering target, not a spending limit. All values are first-principles estimates; nothing is measured.

Safety: These are paper estimates. They do not show that a puck is safe to leave in a roadway, that the primary lithium cells are safe in a hot road, or that the bond will hold. Bond strength, cell temperature and crush behavior must be checked on hardware before any puck is installed, and only under a road permit with traffic management. See LDZ-PRC-001, Safety.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design unless a source is given.

Input	Value	Basis
Traffic	100 state changes a day per puck plus hourly heartbeats: 124 uplinks	LDZ-REQ-001 R3
Payload	11 bytes plus 13 bytes of LoRaWAN overhead, 125 kHz, CR 4/5 (US915 default band; 11 bytes is the most the slowest US915 rate carries)	LDZ-PRC-001; LDZ-DDR-002; LDZ-DEC-001 item 2
Radio currents	45 mA transmit at +14 dBm, 4.6 mA receive, 8 mA controller awake 0.5 s per uplink, two 0.1 s receive windows	Same as FND-CAL-001
Sleep and sampling	6 μ A sleep; one reading a second, 10 ms awake at 2 mA	LDZ-PRC-001 v0.2
Cells	2 x 2.6 Ah in parallel, derated 40 % for cold, pulse loads, self-discharge and end-of-life voltage	ER14505 class, typical rating
Fair use	30 s of uplink airtime and 10 downlinks per node per day on The Things Network	TTN, checked 2026-09-25
Band rules	US915: no duty cycle; a transmission may not exceed 400 ms of dwell time on a 125 kHz channel; uplink rates SF7 to SF10 only (SF11 and SF12 are not offered); the RX2 window is SF12 on 500 kHz	LoRaWAN regional parameters for US902-928; FCC 47 CFR 15.247
Link	Puck antenna -2 dBi once potted; 10 dB road-level loss; 10 dB (typical) to 20 dB (worst) for a van overhead; gateway 10 m high, 2 dBi, 2 dB feeder loss; 6 dB noise figure	Estimates; FND-CAL-001 for the gateway side
Path loss	3GPP TR 38.901 urban microcell street canyon (UMi), Table 7.4.1-1, at 915 MHz with the user height at its 1.5 m floor; the road-level loss covers the lower antenna	Standard model, used outside its height range
Detection	A vehicle is a line of vertical induced dipoles along its length; 10 μ T under the middle of a parked van (calibration); 3 μ T threshold, 10 times the magnetometer's 0.3 μ T RMS noise	Assumption; only a field log can set these
Loads	Half of a 10 t single axle (49 kN), x1.3 dynamic; friction 0.7; rigid cast polyurethane 40 MPa and semi-rigid potting 15 MPa in compression; two-part road-marker epoxy to asphalt shear 1.0 MPa at 20 °C, 0.2 MPa at 50 °C	Typical values, to be checked against chosen materials
Vehicles	Bay 2.6 m wide; van 2.0 m wide, 1.75 m track, 225 mm tires; car 1.8 m, 1.55 m, 200 mm	Typical dimensions
Sign	Class C receiver on at 15.2 mW; 200 redraws a day at 26.4 mW for 5 s plus the controller; 90 % rail efficiency; 7.5 in panel with a 97.9 mm active height, digits 80 % of it	Typical 7.5 in panel data
Legibility	30 ft of distance per inch of letter height (0.36 m per mm)	MUTCD, section 2A.13

■ 2. Airtime (R5)

An 11-byte uplink is on air for 61.7 ms at SF7, 205.8 ms at SF9 and 370.7 ms at SF10, the slowest rate US915 offers on 125 kHz [A1]. At SF10 that is inside the 400 ms dwell limit, which a 12-byte payload (411.6 ms) would break, so the 11-byte rule is what lets the slowest US915 rate work at all [A2]. At 124 uplinks a day the airtime is 7.7 s at SF7, 14.0 s at SF8 and 25.5 s at SF9, all within the 30 s fair-use limit, but 46.0 s at SF10 [A1]. At SF10 even the state changes alone exceed the limit: 81 uplinks a day is the most that fits. SF11 and SF12 are not offered on US915 uplinks, so the 102 s and 184 s of the EU868 figures no longer arise. **R5 is at risk:** met at SF7 to SF9, not met

at SF10. With the 11-byte payload SF9 fits with hourly heartbeats, so the 2 h heartbeat fallback is not needed at SF9. A private gateway (TwinKit) is not bound by the community fair-use limit, and US915 has no duty cycle that would limit it.

■ 3. Energy and cell life (R3)

At SF9 a puck uses 1.11 mAh a day: 0.144 mAh asleep, 0.480 mAh for sampling (20 μ A average) and 0.488 mAh for 124 uplinks at 14.2 mA·s each [B1, B2]. That is 0.406 Ah a year against 3.12 Ah usable, **a life of 7.7 years, and R3 is met** [B2, B3]. Sampling and uplinks each take about 44 % of the budget [B3], so a slower sampling rate is the first lever if life must grow. Life is 9.6 years at SF7 and 6.2 years at SF10 [B2], the slowest US915 rate, so R3 holds at every rate US915 offers, but pucks at SF10 still fail R5.

Pulse supply. One SF9 uplink draws 9.3 mC; if the capacitor supplied it alone with 0.3 V of droop it would need 31 mF, and 56 mF at SF10 [B4]. Bobbin Li-SOCl₂ cells cannot deliver 45 mA pulses well, so the capacitor is essential. Ordinary supercapacitors are often rated only to 60 to 70 °C, below the road-surface target, so the BOM now calls for an 85 °C hybrid pulse capacitor.

Supply voltage. Fresh Li-SOCl₂ cells sit at about 3.67 V open circuit, above the 3.6 V recommended maximum of STM32WL-class modules. One Schottky diode per cell brings this to 3.42 V and also stops one cell charging the other, which primary lithium cells must never do [B5]. This is an engineering proposal (LDZ-DDR-001, O6).

■ 4. Report latency (R2)

A new state must hold for 15 s, sampling adds up to 1 s and the network path about 2 s, so a report reaches the server in 17.7 s typically at SF9 [C1]. US915 has no duty-cycle wait after the previous uplink (the EU868 rule that added 20.4 s at SF9 and gave a worst case of 38.6 s), so the worst case is 18.2 s at SF9 and 18.4 s at SF10 [C1]. A retry on another channel after a lost uplink would add about a second and a half at most [C1]. **R2 is met on paper** at every rate US915 offers; the SF11 and SF12 cases of the EU868 figures cannot occur.

■ 5. Link from road level (R4)

With 11.5 dBm EIRP the link budget at SF9 is 141.0 dB [D1, D2]. At 915 MHz the UMi model gives 127.5 dB of non-line-of-sight path loss at 1 km to a 10 m gateway, and 118.5 dB in line of sight [D1]. Adding 10 dB for an antenna at road level and 10 dB for a van overhead leaves **-6.4 dB at 1 km out of sight and 2.6 dB in line of sight; with the worst vehicle loss, -16.4 dB** [D2]. The range out of sight is 657 m with no fade margin and 342 m with 10 dB [D2]. SF10, the slowest US915 rate, would reach 403 m with the fade margin [D2] but breaks R5. The 1 km target of versions up to LDZ-REQ-001 v0.3 is not met. Under LDZ-DDR-002 (O2) R4 is restated as a gateway within 300 m of the bay; 342 m is 1.14 times that distance [D3], so **R4 is met on paper**, with a thin margin that a field survey must confirm. The UMi model is used below its 1.5 m height floor; the 10 dB road-level loss is the estimate that covers this and only a field survey can settle it.

■ 6. Detection (R1)

With the line-dipole model calibrated to 10 μ T under a parked van, a car gives 10.6 μ T, a small car pushed to one end of the 7 m slot 10.5 μ T, a high-chassis box truck 7.7 μ T and a box truck with little steel low down 3.85 μ T, just 1.28 times the 3 μ T threshold [E1]. A van passing in the next lane gives 0.46 μ T and a van parked in the next slot 0.08 μ T, both well below the threshold [E1]. The geometry therefore favors the slot-center position (D3): vehicles

anywhere in the slot are seen, and neighbors are not. The margins rest on the 10 μ T calibration, which is an assumption. **R1 is at risk and not verifiable at TRL 3**; high-clearance trucks are the weak case, and a short field log under parked vehicles would settle it.

■ 7. Loads and bond (R6)

The load basis is now set: half of a 10 t single axle, 49.0 kN static, or 63.8 kN with a 1.3 dynamic factor for slow maneuvering [F1]. If the whole wheel bears on the 92 mm flat crown, the stress is 7.38 MPa static and 9.59 MPa dynamic; spread over the base it is 2.78 MPa [F1]. That gives factors of 5.4 (static) and 4.2 (dynamic) on rigid cast polyurethane, but only 2.0 and 1.6 on the potting under the crown [F2]. The bond is weaker: with a friction coefficient of 0.7 the tire can push 34.3 kN sideways, 1.51 MPa over the 170 mm pad, against an assumed adhesive-to-asphalt strength of 1.0 MPa at 20 °C (factor 0.66) and 0.2 MPa at 50 °C (0.13) [F3]. The bed is now two-part road-marker epoxy rather than a bitumen pad (LDZ-DDR-002, O3), so the 1.0 MPa figure is the epoxy-to-asphalt value, in which the asphalt surface usually governs. These are upper bounds, since a truck tire deflects around a 31 mm puck and rarely brakes on it. Wheel geometry helps: a van or car parked inside the bay lines keeps its nearest tire edge at least 462 mm (van) or 275 mm (car) from the puck axis, against a 75 mm puck radius, so parked wheels straddle the puck and only maneuvering wheels cross it [F4]. R6 is restated with this finding (LDZ-DDR-002, O3): the load case is a maneuvering wheel crossing the puck, and the slot-center placement keeps it out of parked wheel paths. **R6 is still at risk** on debonding under braking; crushing is met on paper. The pull-off and shear test on asphalt that would settle it is TRL 4 work and on hold.

■ 8. Size and mass (R7)

The model gives a puck 31.0 mm high on a 150 mm base, a 170 mm pad, a 104 mm crown and a 6 mm crown radius [G1]. **R7 is met**. From the model volumes the puck weighs 451 g without its pad, and the pad 75 g [G2]; the TRL 2 estimate of 0.35 kg is corrected, because the cavity is now fully potted.

■ 9. Sign option (R11, R12)

Power (R12). The class C receiver draws 15.2 mW, or 0.364 Wh a day, and 200 redraws add 0.015 Wh; with 90 % rail efficiency the sign draws 0.424 Wh a day, or 17.7 mW [H1]. That is 18 % of FieldNode's allowance of 100 mW, which FieldNode now publishes (FND-DDR-002) [H2]. **R12 is met on paper**. The sign takes about 200 downlinks a day, against The Things Network's limit of 10 [H5], so D7 works only on a private network server such as TwinKit, where the downlinks use 58 s a day in the US915 RX2 window (SF12 on 500 kHz, 289 ms each), with no duty-cycle limit [H5]. This is decided: the sign option requires a private network server or a city network that allows it (LDZ-DDR-002, O5). Typical 7.5 in e-paper panels are rated for about 0 to 50 °C in operation, so the sign may not refresh on freezing days; this is a limit to confirm with the panel chosen.

Legibility (R11). Digits 78 mm high on the 97.9 mm panel read at about 28.2 m by day at full contrast, using the MUTCD legibility index of 0.36 m per mm; 69 mm digits would be enough for 25 m [H4]. E-paper contrast is lower than a printed sign's, so the daytime figure is optimistic. The panel is unlit, and a 0.5 W front light for 12 h a night would add 6.0 Wh a day, 2.5 times the 100 mW allowance [H3]. Under LDZ-DDR-002 (O4) R11 is restated as a daylight-only target of 25 m, so **R11 is met on paper**; at night drivers rely on the data feed.

■ 10. Cost (R13)

Value-engineering target: USD 120 (`budget_usd`). Estimated cost of the constructable design: **USD 116.00 for the two-puck kit (USD 4.00 under the target)**; one puck costs USD 54.00 [I1]. The target covers the two-puck kit only (D6). Making the design constructable added USD 2.00 for sealant, screws and mould release (LDZ-DDR-003). The sign option costs USD 101.00 (USD 97.00 before its channel brackets), and its host FieldNode USD 139.50 (the base node, priced in the FieldNode BOM and up from USD 126.00 in version 0.3), for USD 356.50 per bay with the sign; the one-off dome casting tooling, USD 35.00, is not a per-kit cost [I1]. **R13 is met on paper (under the target)**, on indicative prices.

■ 11. Requirement status

Table 2. Requirement status from this note. The same rows are in `docs/04-calcs/results.csv` .

ID	Requirement	Value	Target	Status
R1	Detect slot state	Van 10.0 μ T, high-chassis truck 7.7 μ T, weak-steel truck 3.85 μ T, next lane 0.46 μ T against 3 μ T	97 % correct over a day	At risk (not verifiable at TRL 3)
R5	Fair-use airtime	7.7 s at SF7, 25.5 s at SF9, 46.0 s at SF10 (SF11 and SF12 not offered in US915)	30 s a day or less	At risk (not met at SF10)
R6	Survive traffic	Crown 7.4 MPa, factor 5.4 on PU and 2.0 on potting; epoxy bond shear 1.51 MPa, factor 0.66 at 20 °C	49 kN maneuvering wheel crossing the puck, no cracking or debonding; parked wheels straddle it	At risk (debond under braking)
R14	Quick install	Bonded pad, no road cutting	15 min per puck	Not verifiable at TRL 3
R2	Report quickly	17.7 s typical; 18.2 s worst at SF9, 18.4 s at SF10 (no duty-cycle wait in US915)	60 s or less	Met on paper
R3	Battery life	7.7 years at SF9; 6.2 years at SF10	5 years or more	Met on paper
R4	Reach a gateway from road level	342 m NLOS range with a van over and 10 dB fade margin at SF9	Gateway within 300 m in a street canyon (restated)	Met on paper
R7	Low profile	31 mm high, 150 mm diameter, 6 mm crown radius, yellow	35 mm or less, rounded	Met on paper
R11	Sign legible (option)	About 28 m by day at full contrast; unlit at night	25 m by day (restated, daylight only)	Met on paper
R12	Sign power (option)	0.42 Wh a day (17.7 mW) against 100 mW	Within the FieldNode allowance	Met on paper (private network only)
R13	Low cost	USD 116.00 for two pucks, USD 4.00 under the target	Value-engineering target USD 120	Met on paper (under the target)
R8	Weather and temperature	Parts rated -40 to +85 °C with an 85 °C pulse capacitor; fully potted	IP68; -25 to +70 °C	Met by design (sealing not verifiable at TRL 3)
R9	Privacy	Magnetometer only; state, timer, confidence, voltage, temperature	No images, audio or identifiers	Met by design
R10	Open data	CDS Events <code>park_start</code> , <code>park_end</code> , <code>scheduled_report</code> , <code>comms_lost</code> , <code>comms_restored</code> ; Metrics for occupancy and dwell	Maps onto CDS Events and Metrics	Met by design
R15	Tamper resistance	No external fasteners; removal flagged by a field and orientation step	No screws; removal reported	Met by design

Counts: none not met, 3 at risk, 1 not verifiable at TRL 3, 7 met on paper, 4 met by design [K]. In version 0.1: 2 not met (R4, R11), 3 at risk, 1 not verifiable, 5 met on paper, 4 met by design.

The Curb Data Specification event types used for R10 were checked against the specification's Events API on 2026-09-25 ([Open Mobility Foundation, CDS Events](#)).

■ 12. Checks against the TRL 2 figures

Table 3. TRL 2 claims (LDZ-PRC-001 v0.2) against this note.

TRL 2 claim	This note	Action
About 20 s to report	17.7 s typical, 18.2 s worst at SF9 (US915)	Precis updated
About 1.2 mAh a day, about 7 years	1.11 mAh a day, 7.7 years at SF9	Precis updated
72 ms at SF7, 247 ms at SF9; 9 s and 31 s a day	61.7 ms and 205.8 ms for 11 bytes; 7.7 s and 25.5 s a day	Corrected; R5 now met at SF9
A vehicle overhead leaves room for a gateway a few hundred meters away	342 m with a 10 dB fade margin	Stands; R4 restated to 300 m and met on paper (LDZ-DDR-002)
2.8 MPa over the base; printed housing would fail	2.78 MPa over the base, 7.38 MPa at the crown; cast PU adopted	Precis updated; bond is the new risk
Sign legible at about 10 m	About 28 m by day at full contrast	Corrected; R11 restated to daylight only and met on paper (LDZ-DDR-002)
Sign about 0.4 Wh a day	0.42 Wh a day drawn	Stands
Puck about 0.35 kg	451 g without pad	Corrected
Two-puck kit about \$110	USD 116.00 with the diodes, 85 °C capacitor and construction consumables	Corrected